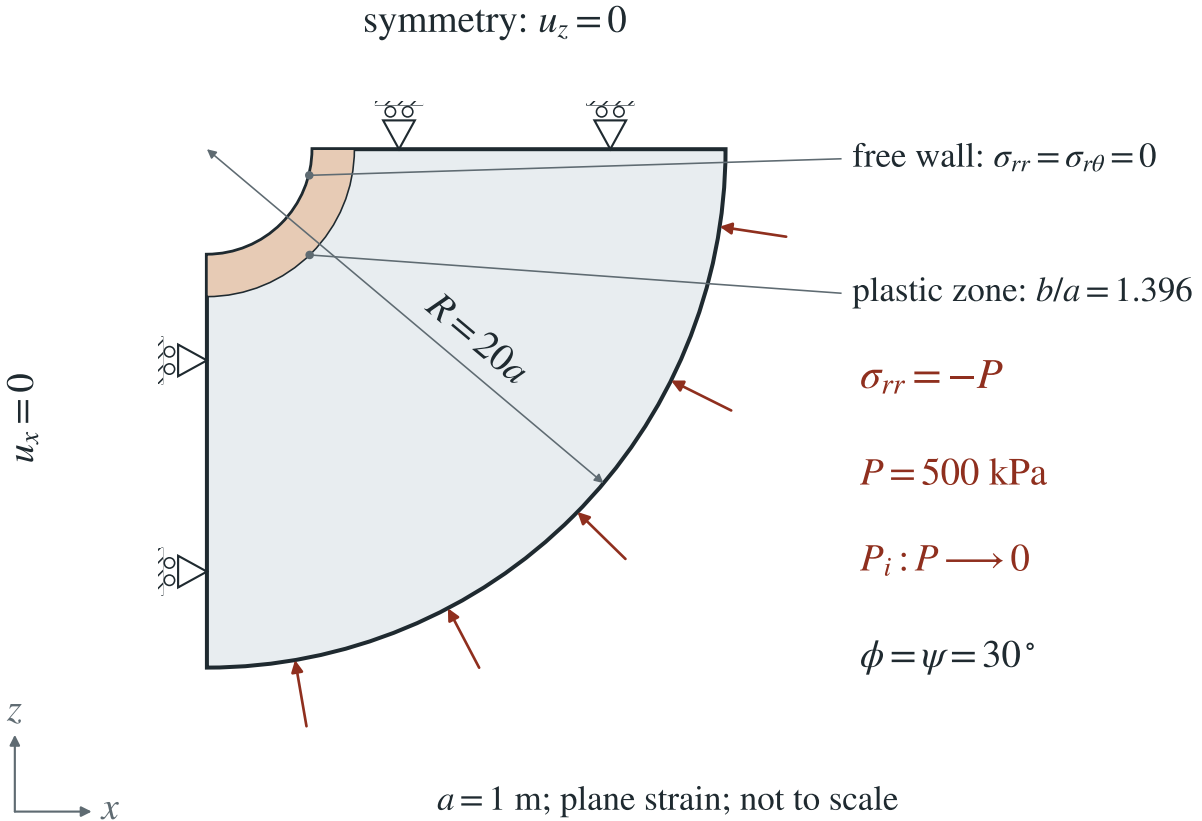
symmetry: $u_z = 0$
free wall: $\sigma_{rr} = \sigma_{r\theta} = 0$
plastic zone: $b/a = 1.396$
$R = 20a$
$\sigma_{rr} = -P$
$P = 500$ kPa
$P_i : P \longrightarrow 0$
$\phi = \psi = 30^\circ$
$u_x = 0$
z
x
$a = 1$ m; plane strain; not to scale